

NOP (00)			Cycles: 2	Size: 1
Implied				
Cycle	R/W		Desc.	
-X.2	Read PC		Store as OpCode.	
1.1			Inc. PC .	
1.2				
2.1				
2.2	Read PC		Store as OpCode.	
+X.1				

TCALL (01)	Cycles: 8	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
4.1		
4.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
5.1		Dec. SP by 2.
5.2		
6.1		
6.2	Read \$FFDE	Store as PC .L.
7.1		
7.2	Read \$FFDF	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

SET1 (02)	Cycles: 4	Size: 2
Direct Page Bit (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~1) 1.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

BBS (03)	Cycles: 5-7	Size: 3
Direct Page Bit Relative		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 0)) == (1 << 0).
3.2		
4.1		
4.2	Read PC	Store as Operand.
5.1		Inc. PC . If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = PC + int8(Operand). PC .L = Address.L.
6.2		
7.1		PC .H = Address.H.
7.2	Read PC	Store as OpCode.
+X.1		

OR (04)	Cycles: 3	Size: 2
Direct Page (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
3.2	Read PC	Store as OpCode.
+X.1		

OR (05)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

OR (06)	Cycles: 3	Size: 1
Indirect X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1	Read (X ((PSW & P flag) << 3))	Inc. PC .
1.2		
2.1		
2.2		Store as Operand.
3.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
3.2		Store as OpCode.
+X.1		

OR (07)	Cycles: 6	Size: 2
X-Indexed Indirect (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Store as Pointer.H.
5.1		
5.2	Read Pointer	Store as Operand.
6.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

OR (08)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . A = Operand. N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		

OR (09)	Cycles: 6	Size: 3
Direct Page to Direct Page (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand0.
3.1		
3.2	Read PC	Store as Operand.
4.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
4.2	Read Address	Store as Operand1.
5.1		Operand0 = Operand1. N flag = (Operand0 & \$80), Z flag = !Operand0.
5.2	Write to Address	Write Operand0.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

OR1 (0A)	Cycles: 5	Size: 3
Absolute Boolean Bit		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read (Address & \$1FFF)	Store as Operand.
4.1		Bit = Address >> 13. C flag = ((Operand & (1 << Bit)) != 0).
4.2		
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

ASL (0B)	Cycles: 4	Size: 2
Direct Page (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		C flag = (Operand & \$80). Operand <<= 1. N flag = (Operand & \$80), Z flag = !Operand.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

ASL (0C)	Cycles: 5	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

PUSH (0D)		Cycles: 4	Size: 1
Implied			
Cycle	R/W		Desc.
-X.2	Read PC		Store as OpCode.
1.1			Inc. PC .
1.2	Read PC		Discard.
2.1			
2.2	Write to u8(SP +0) \$0100		Push PSW onto stack.
3.1			Dec. SP by 1.
3.2			
4.1			
4.2	Read PC		Store as OpCode.
+X.1			

TSET1 (0E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		Tmp = (A - Operand). Operand = A . Z flag = (Tmp == 0), N flag = (Tmp & \$80).
4.2	Read Address	Discard.
5.1		
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

BRK (0F)	Cycles: 8	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		
2.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
3.1		
3.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
4.1		
4.2	Write to u8(SP -2) \$0100	Push PSW onto stack.
5.1		Dec. SP by 3. I flag = 0, X flag = 1.
5.2		
6.1		
6.2	Read \$FFDE	Store as PC .L.
7.1		
7.2	Read \$FFDF	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

BPL (10)	Cycles: 2-4	Size: 2
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Set Jump to (PSW & \$80) == \$00.
1.2		Store as Operand.
2.1		Inc. PC . If !Jump, end instruction (next half-cycle is 4.2).
2.2		
3.1		Address = PC + int8(Operand).
		PC .L = Address.L.
3.2		
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

TCALL1 (11) Cycles: 8		Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
4.1		
4.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
5.1		Dec. SP by 2.
5.2		
6.1		
6.2	Read \$FFDC	Store as PC .L.
7.1		
7.2	Read \$FFDD	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

CLR1 (12)	Cycles: 4	Size: 2
Direct Page Bit (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~1) 0.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

BBC0 (13)	Cycles: 5-7	Size: 3
Direct Page Bit Relative		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 0)) == (0 << 0).
3.2		
4.1		
4.2	Read PC	Store as Operand.
5.1		Inc. PC . If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = PC + int8(Operand). PC .L = Address.L.
6.2		
7.1		PC .H = Address.H.
7.2	Read PC	Store as OpCode.
+X.1		

OR (14)	Cycles: 4	Size: 2
X-Indexed Direct Page (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2	Read Address	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		